Causal Inference using Machine Learning for Agricultural Policy Decisions

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Prepared for
ICAR-NIAP Winter School: Advanced Tools for Impact Assessment of Agricultural Technologies and Policy Analysis, New Delhi, India

Who am I?

- Spatial Economist at CIMMYT, New Delhi Office
  - Previously: World Bank Consultant (2017-2021), Statistical computing lecturer, Bunda College (2010-2013).
- **Methodological areas of interest:** Bayesian and spatial econometrics, risk modeling, mathematical programming, *applied causal machine learning*.
- **Agricultural areas of interest:** Economics of agricultural research, technology adoption, agricultural risk management, regional agricultural trade.
- Born and raised in **Malawi**, a small country in Southern and Eastern part of Africa.
- Education:
  - BSc in Agricultural Economics (2009), Bunda College, Malawi
  - MSc in Research Methods (2012), JKUAT, Kenya;
  - MSc and PhD in Applied Economics (2015, 2019), University of Minnesota, USA

Outline

- Craft of causal econometrics for agricultural policy analysis: Recap
- Craft of machine learning: Overview
- Causal ML algorithms: Primer and selected examples
- Heterogenous treatment effects and causal random forests
- Applications
- Policy learning

Slides and reading materials:
https://github.com/MaxwellMkondiwa/Training_Materials/tree/main/2026/NIAP

Session 1: Causal Inference recap and ML primer

Big data requires Factology! Causal Inference

Agricultural economists and other social scientists **[Researchers]** tend to take **data as facts**.

The problem is the **data are not facts**. Facts are what is really there. Data are quantitative representation of facts, which statistical workers and economists concoct.

I call the study of how primary statistical information is made into economic data **"factology."** The **neglect of factology risks scientific ruin**.

The bottom line is, for data producers, that **full disclosure of procedures and labeling of data series are essential**; and, for **data users,** be careful and investigate the **data before using them**.

Bruce Gardner (1992, AAEA fellows lecture)

5 cultures framework for agricultural policy analysis

1.Causal inference (Experiments, quasi-experimental methods)

2. Prediction (Random forest, neural networks)

3. Causal and Knowledge guided machine learning (1+2)

4. Decision support and recommender systems (3+ policy rule)

5. Equilibrium/ system optimization/process understanding (Positive mathematical programming, agent based models)

Statistical Science
2001, Vol. 16, No. 3, 199–231

Statistical Modeling: The Two Cultures

Leo Breiman

Harvard Data Science Review • Issue 5.1, Winter 2023

Statistical Modeling: The Three Cultures

Adel Daoud[1,2] Devdatt Dubhashi[2]

Breiman's Two Cultures: A Perspective from Econometrics

Guido Imbens, Susan Athey

Observational Studies, Volume 7, Issue 1, 2021, pp. 127-133 (Article)

Wiley Online Library Search

ECONOMETRICA
JOURNAL OF THE ECONOMETRIC SOCIETY

Original Articles

Policy Learning With Observational Data

Susan Athey Stefan Wager

First published: 15 January 2021 | https://doi.org/10.3982/ECTA15732 | Citations: 51

Mkondiwa et al 2023: Lead lecture at Haryana Agricultural University Conference

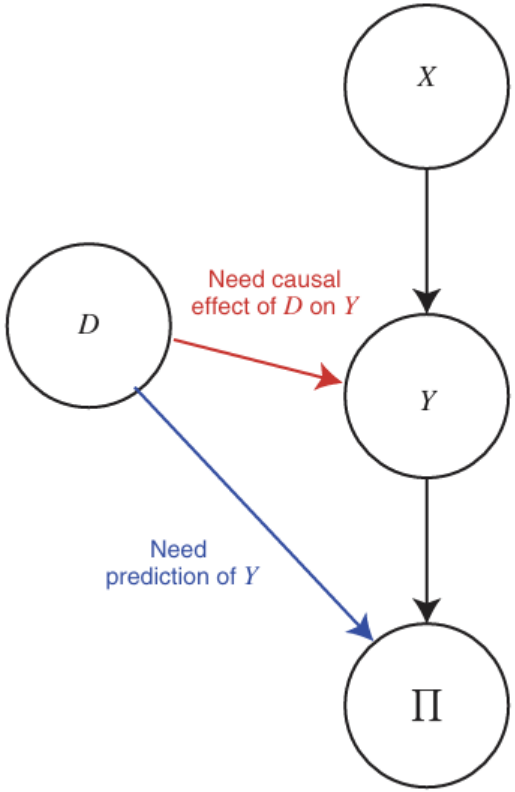
Culture 1: Causal Inference

- Credibility revolution in economics.
- Use of randomized control trials and quasi-experimental methods is now the norm.
  - You have probably already discussed these at length
- No more **"kitchen sink regressions"** in serious agricultural economics journals
- **Recall:** Sources of *statistical* endogeneity $Cov(x, \epsilon) \neq 0$: Reverse causality, omitted variables, and measurement error.

Should we all do causal policy analysis only?

No according to Mullainathan (2025 AEA Distinguished Lecture)

- See causal graph
- If decision D has potential effect on Y, then we need causal estimate
  - What is effect of new wheat variety A on yield? Profits?....
- If decision D only affects payoffs Π, then we need to predict Y
  - How many farmers are food insecure?



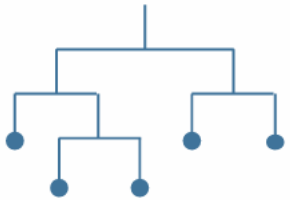
Culture 2: Predictive modeling
(Yes, some policy questions are purely predictive)

Predictive agricultural policy question

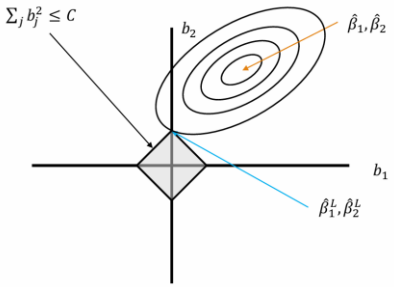
- What is the crop price in some unsampled village X?
- How many farmers are structurally poor? Or food insecure?
- What proportion of women farmers are empowered in state X?

Machine learning tools used

- Random forest to predict prices at grid level (Madaga et al 2024)
- Tennant et al (2024 PNAS), McBride et al 2021 (AEPP)



- LASSO (Saha and Narayanan 2022 World Development).



Session 2: Causal AI

- Causal ML algorithms: Primer
- Heterogenous Treatment Effects and Causal Random Forests

ML. What is it good for? Prediction

- Collapse high-dimensional data down to something useful. Don't care that much how it happens (if you care there is explainable ML/interpretable ML).
  - Collapse pixels to single classification
  - Collapse text down to sentiment or score
  - Reduce large number of regressors to small and correct number
- What if we want to predict a counterfactual? Important for causal inference
- We have many ways to this
  - RCT, regression controls, similar untreated entities via matching, Quasi-exp techniques, simulation via structural models, synthetic controls.

Adapted from Steve Miller Slides

Culture 3: What is (causal) AI?

- **Predictive modeling has major limitations.**
  - Black box. You don't know why it works so well.
  - Not ideal for new sites and contexts.
- **Causal ML:** Combines causality research (cause-effect) and predictive modeling (AI).
  - Cause-effect represents theory or scientific basis.
  - Also called **Knowledge Guided ML (KGML), physics-aware, hydrology-guided, e.t.c**
  - **In economics, mostly called causal ML (Athey and others).**
- **Main goal:** Produce heterogenous treatment effects (HTEs) in the potential outcomes framework called conditional average treatment effects (CATEs) **(Athey et al 2019).**

$$\tau_i = Y(1) - Y(0)|X$$

Augmented inverse probability estimator (AIPW)

But why study heterogenous policy effects?

- Policies do not have uniform effects. They vary:
  - **Spatially**
  - **By firm**
  - **By individual**

- **Can also vary by duration, time,…**
  - Form of heterogeneity may be useful in targeting future (changes to) policies

What this algorithm achieves? Emerging research paradigm on personalized advisories

News & Insights

Machine-learning technique identifies people who would benefit most from treatment to reduce future cardiovascular disease risk

Inoue, Athey, Tsugawa 2023

April 13, 2023
By Enrique Rivero
5 min read

New UCLA research suggests that a novel machine-learning technique known as "causal forest" was about five times more efficient than the current clinical practice of treating patients with high blood pressure.

Under current practice, physicians treat patients with high blood pressure under the assumption that people at greatest risk for adverse outcomes and death from a disease benefit the most from lowering blood pressure. In recent years, however, there have been advances in how machine-learning can predict adverse health outcomes due to high blood pressure based on one's individual characteristics, which allows clinicians to examine the extent to which an intervention's effects can vary between individuals and thus tailor treatments to

Flickr: Marco Verch Professional Photographer

Source: https://www.uclahealth.org/news/machine-learning-technique-identifies-people-who-would

Personalized medicine: Which form of cancer therapy is most appropriate for this specific patient?

Targeted marketing/advertising/offers: What is the best offer for this individual

Targeted agricultural extension approach??

Targeted (bundles of) agronomic practices??

But we can study heterogeneity in other ways?

Yes

- Estimate models on subsamples
- Interaction terms in models
- Mixed effects or random effects models
- Regression trees/Random forests

But

- Which subsamples should we choose?
- Which interaction terms do we include?
- Is the chosen functional form right? Or parametric assumption right?
- Predictive tools are not causal

Modeling challenge

- Single treatment effect τ_0 model

$$y_i = \beta_0 + \tau_0 T_i + \gamma X_i + \mu_i$$

- Lots of specifications for heterogenous treatment effects

$$y_i = \beta_0 + \tau_0 T_i + \tau_{xi} X_{i1} T_i + \gamma X_i + \mu_i$$

- In general, we can write these models as

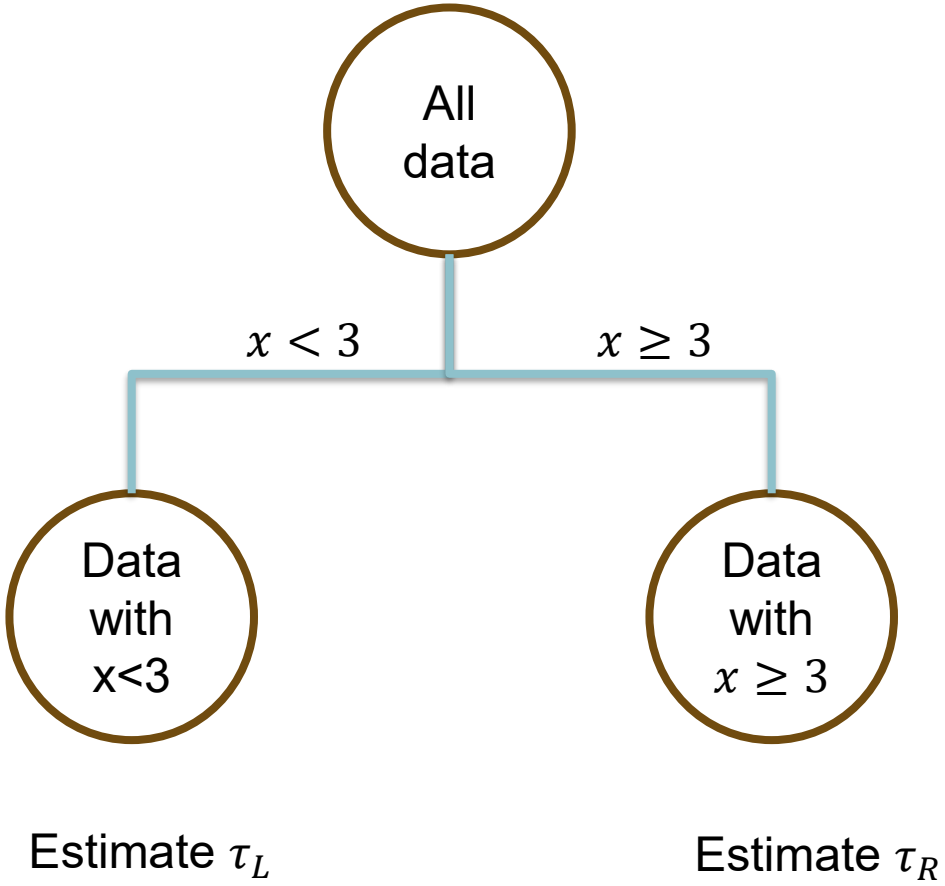
$$y_i = \beta_0 + T_i \tau(X_i) + \gamma X_i + \mu_i$$

Where $\tau(X_i)$ represents the treatment effect for a unit with characteristics X_i

What should we choose for $\tau(x)$?

- Conventional wisdom
  - Use theory
  - Try few specifications to test different hypotheses
  - Maybe use some specification tests
- Athey and Imbens
  - Assume we have controlled for enough (ignorability)
  - Let a regression tree decide the form of $\tau(x)$

Idea, graphically



- Core estimation in each branch is still using econometric methods, assumptions.
- Tree just helps us find subsamples where the treatment effect differs most across subsamples
- When is this ok?
- If we have unbiased estimators of τ_L and τ_R

Causal tree method

- Goal: Use trees to predict treatment effect $\tau(x)$, not outcome y
- Say we observed $\tau(x)$. Just let tree predict $\tau(x)$, splitting on predictors x.
- Key challenge: We don't observe $\tau(x)$
- Idea: estimate $\tau(x)$ with $\hat{\tau}(x)$ and predict $\hat{\tau}(x)$ still splitting on x
  - Assume conditional un-confoundedness/ ignorability
  - Use inverse propensity score weighting estimator within branches
  - Work out splitting criterion, cross validation criterion, SE estimation

Causal tree intuition

- Plain old regression trees split the sample to minimize squared prediction error

$$\sum_{i:left} (y_i - \hat{y}_i)^2 + \sum_{i:right} (y_i - \hat{y}_i)^2$$

- To predict treatment effects, we are more interested in minimizing

$$\sum_{i:left} (\tau_i - \hat{\tau}_i)^2 + \sum_{i:right} (\tau_i - \hat{\tau}_i)^2$$

- In most cases, this is equivalent to maximizing the variance of predicted treatment effects $\hat{\tau}_i$ across leaves. Why? Predictions discriminate more
- Choose splits to maximize $var(\hat{\tau}_i)$

Model

- Potential outcomes (PO) model

$$\boldsymbol{\tau(x) = [Y_i(1) - Y_i(0) | X_i = x]}$$

- If using OLS, τ may be biased because of selection bias, omitted variable bias, measurement error, and reverse causality

Augmented Inverse Probability Weighted Estimator (Athey et al 2019)

- AIPW estimator aka Double Robust Estimator

$$\hat{\Gamma}_i = \hat{\mu}_1(X_i) - \hat{\mu}_0(X_i) + \frac{W_i}{\hat{e}(X_i)}\left(Y_i - \mu_1(X_i)\right) - \frac{1 - W_i}{1 - \hat{e}(X_i)}\left(Y_i - \hat{\mu}_o(X_i)\right)$$

- $\hat{e}(X_i)$: Probability of being treated
- $\hat{\mu}_1(X_i) - \hat{\mu}_0(X_i)$**:** Conditional outcome component; meta-learner. Robust 1
- **The other terms:** Inverse Probability Weighted (IPW) Estimator. Robust 2
- **AIPW advantage:** Either the exposure or outcome equation need to be correct

- **RandomForest:** Minimize Mean Squared Error (MSE) in each leaf for prediction.
  - Optimized for yield prediction.

- **Causal RF:** Minimize MSE in each leaf for heterogeneous treatment estimation (difference between outcome of treatment and control).
  - Optimized to detect heterogeneous treatment effects

Why causal random forest?

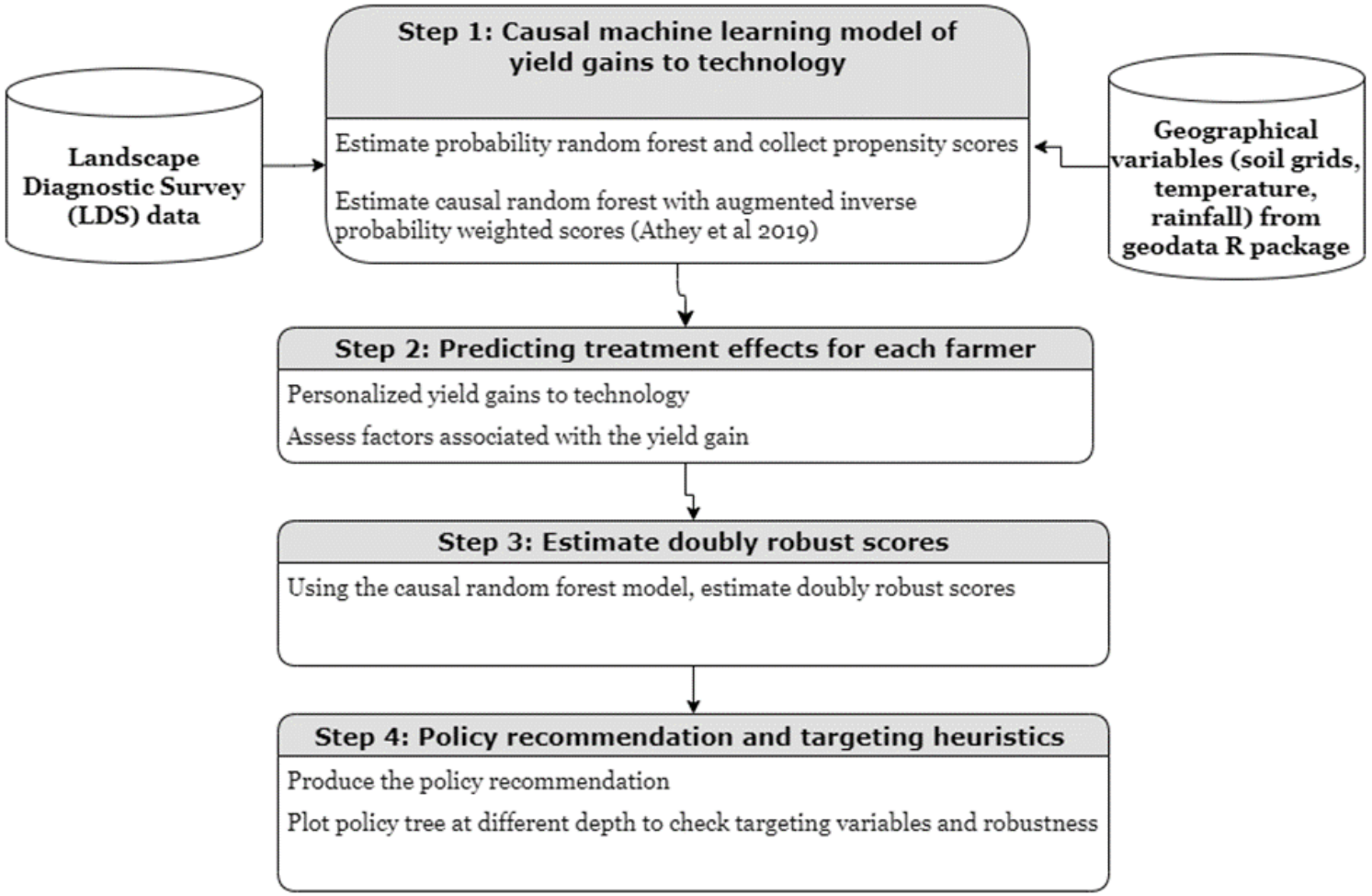
- Some people wrongly use a Random Forest to simply learn two predictions and take the difference

$$\tau^{RF}(x) = \underbrace{\{\hat{y}^{(RF)}(W = 1, X = x) - \hat{y}^{(RF)}(W = 0, X = x)\}}_{Wrong}$$

- Even under an RCT, $\hat{\tau}^{RF}(x)$ will be biased
- **Why?** Because we used the same data to decide how to partition into leaves and to estimate mean values within each leaf.
- Causal trees help resolve this problem using **Honest splitting.**

- **Honest splitting:** Splitting using training sample and uses independent sample to estimate leaf means.

Workflow diagram



https://maxwellmkondiwa.github.io/spatial_exante_sop_book/

https://eia2030-ex-ante.github.io/causal_RF_targeting/

Example: Personalized agronomic recommendations to smallholder farmers

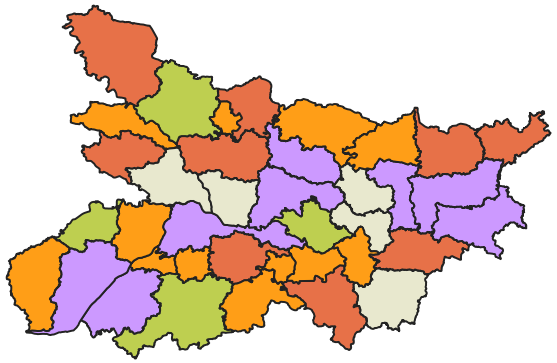
- Propose Causal ML (Athey et al 2019) and policy tree optimization (Athey and Wager 2021) as **an integrative framework for generating spatially explicit and personalized crop management recommendations**

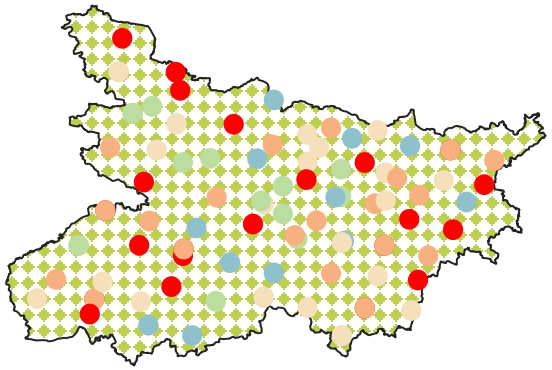
  - **Goal:** Find and evaluate a minimal set of variables (3-4) to use for targeting out-of-sample -> technology options that give maximum returns to individual advisory recipients.

- Extends **traditional parametric production function and optimization framework** to handle complex farmer heterogeneity and variable interactions.

Trajectory of data-driven agricultural recommendations

Blanket recommendations → Spatially-targeted recommendations → Contextually-targeted recommendations





Desired features:

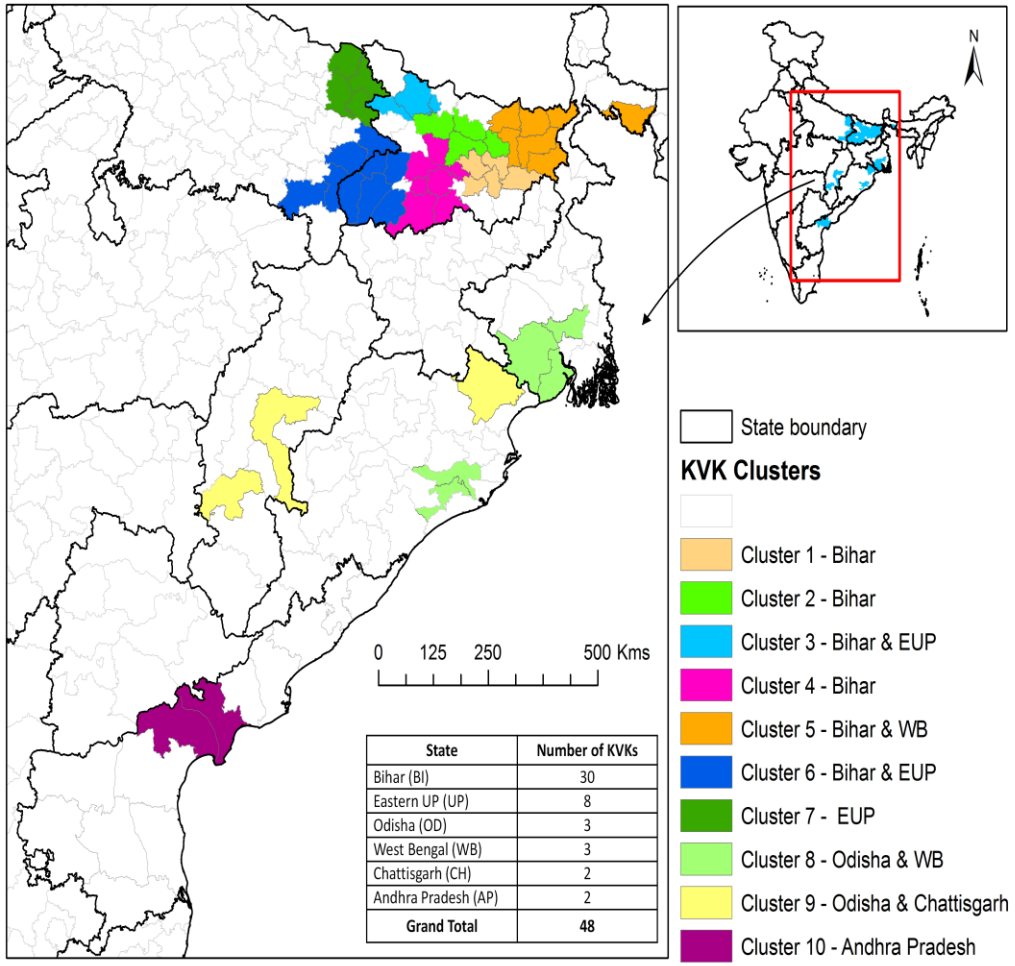
- Bias free estimates of returns to adoption
- Profit & risk incorporated into objective function
- Parsimonious predictions for out of sample recommendation

What does this have to do with ***contextual impact assessment****?*

CSISA-ICAR-KVK LDS geo-coded data

- **Landscape Diagnostic Survey** data (>45,000 landscape crop assessments especially for wheat and rice)
- **Multi-topic crop assessment**
  - Basic demographic variables
  - Landholding
  - Crop establishment
  - Fertilizers
  - Irrigation
  - Weeding
  - Pests and diseases
  - Other shocks e.g., flooding, droughts
  - Harvesting and yield

- **Complemented by large N on-farm field trials (with KVKs) and RCT.**



State	Number of KVKs
Bihar (BI)	30
Eastern UP (UP)	8
Odisha (OD)	3
West Bengal (WB)	3
Chattisgarh (CH)	2
Andhra Pradesh (AP)	2
Grand Total	**48**

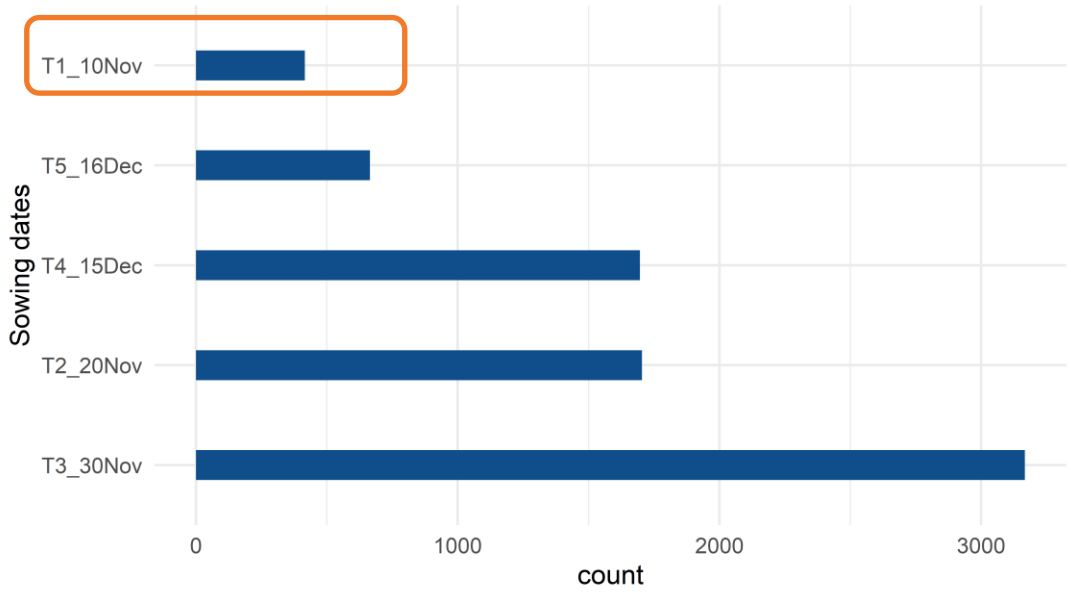
LDS data available at: https://data.cimmyt.org/dataverse/csisadvn

Data and Context

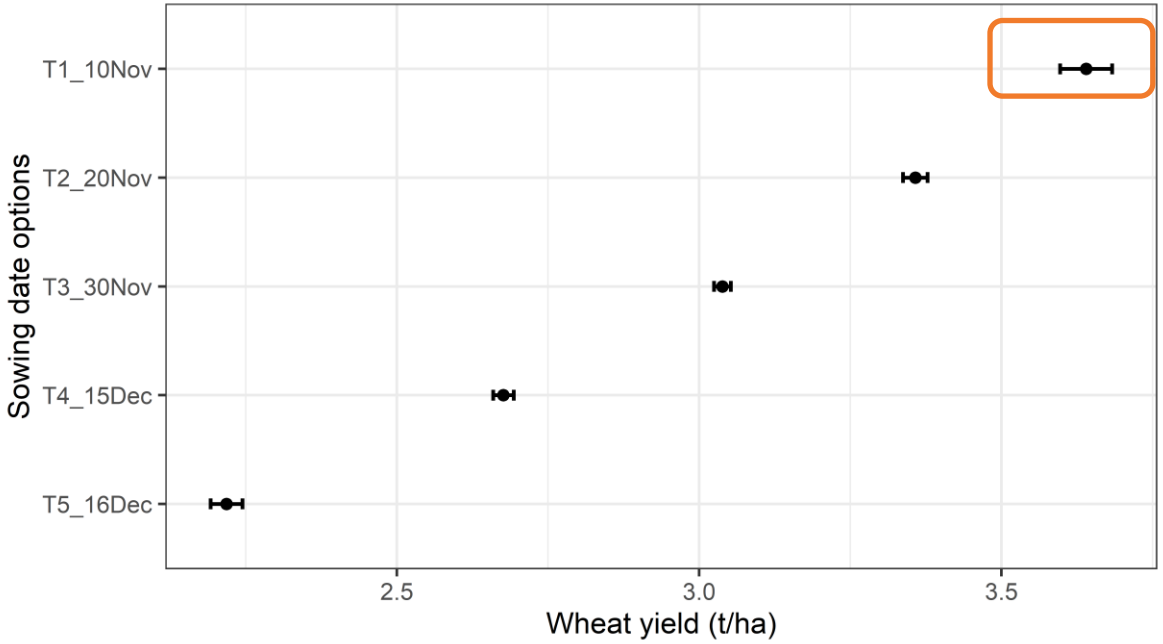
- 2017-19 CSISA Landscape Diagnostic Survey (LDS) ~ 8000 wheat farms in Bihar
- **Outcome variable:** Wheat yield (t/ha)
- **Weed management**
  - None, Manual, Manual + herb., 2-4D, Sulfosulfuron + metsulfuron
- **Sowing date:** Following prior agronomic trials, we use five sowing schedules
  - T1: before 10th Nov
  - T2: 10-20th Nov
  - T3: 20-30th Nov
  - T4: 30-15th Dec
  - T5: After 16th Dec
- **Varietial choice:** Use 12 varieties accounting for 82.73% of wheat varieties grown
- **Irrigation management:** 1,2,3,4 times
- **Controls:** Shocks (drought, weeds, insects), Genotype (Maturity class), Environment (Slope, ruggedness, temp, precipitation, elevation, latitude, longitude, soil N, soil sand, soil pH, soil OC),Management (N, P, Irrigation, Sowing, Weed management), Socioeconomics (Gender, Market distance)

Example: Wheat sowing dates mismatch

Farmers sowing in the window

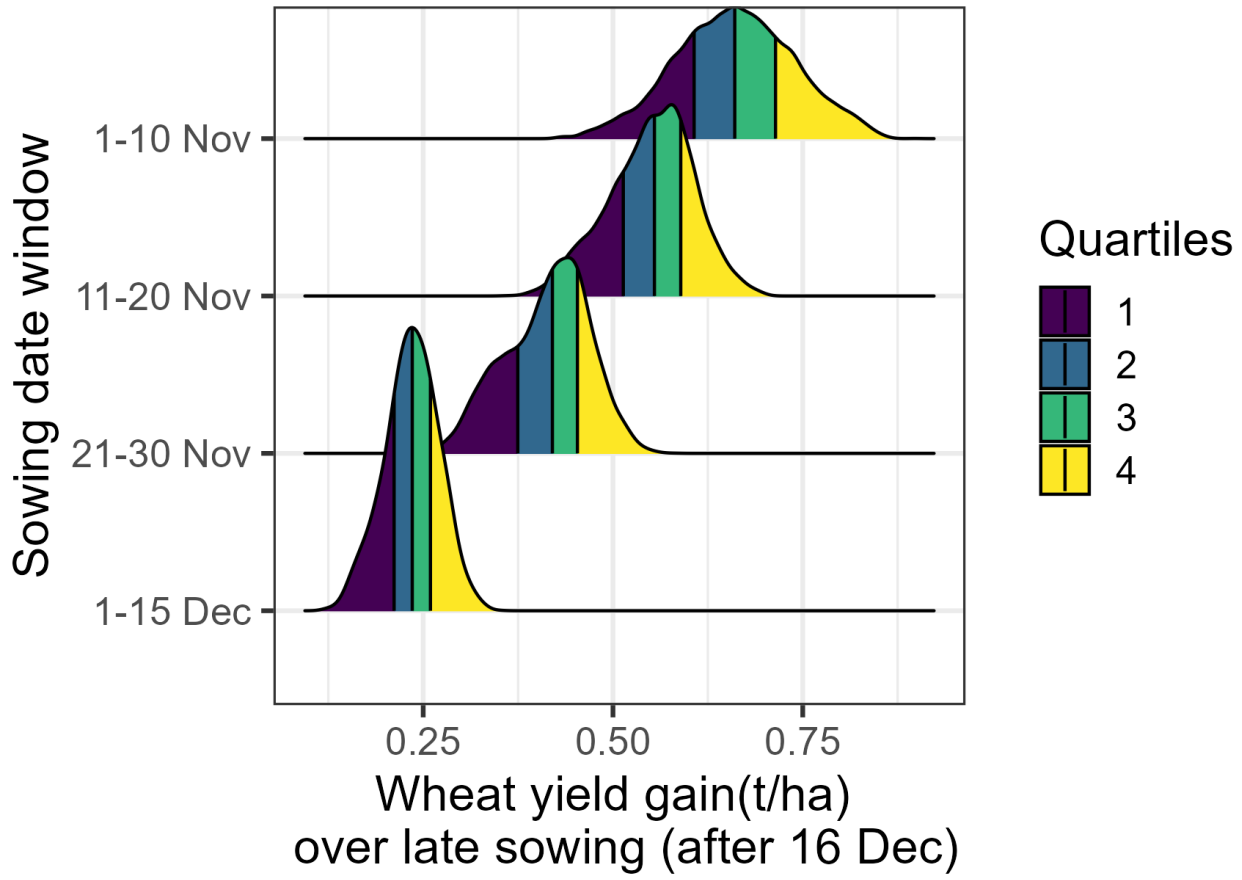


Average wheat yields

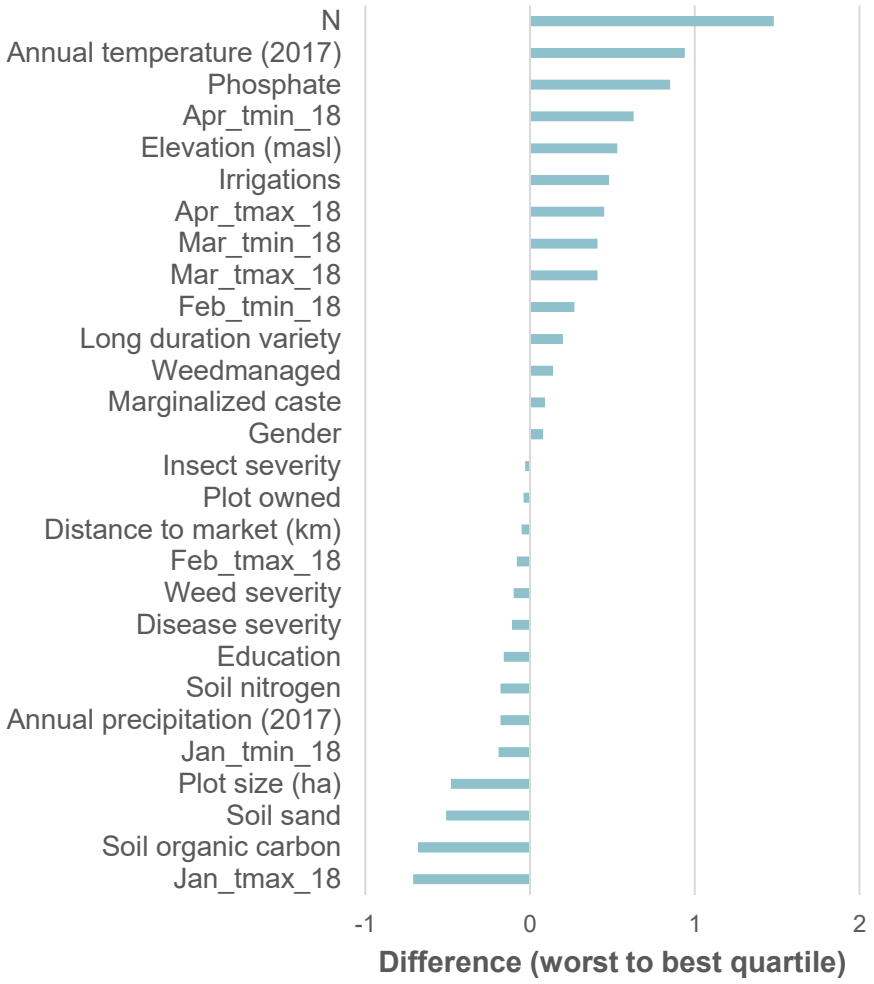


Heterogeneity of expected returns if sowing in any of windows

Causal RF results: Personalized yield gains to early sowing

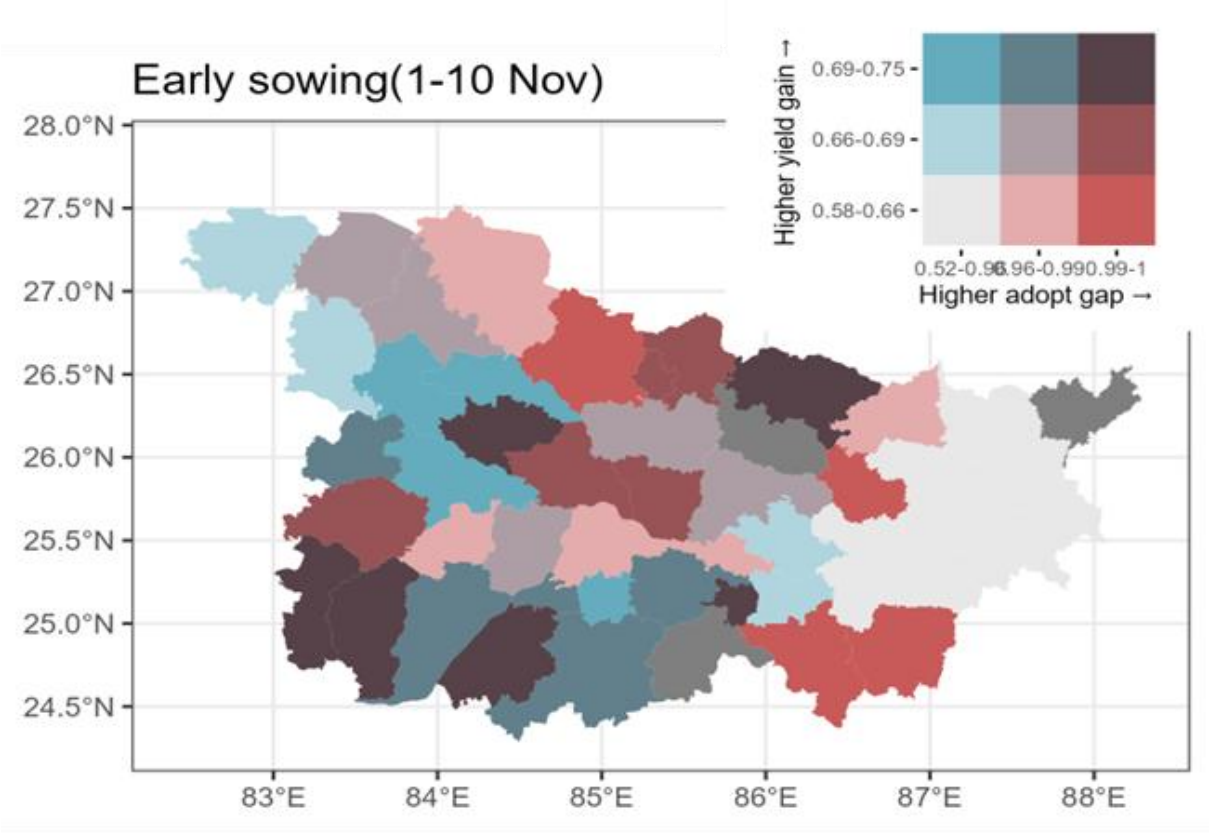


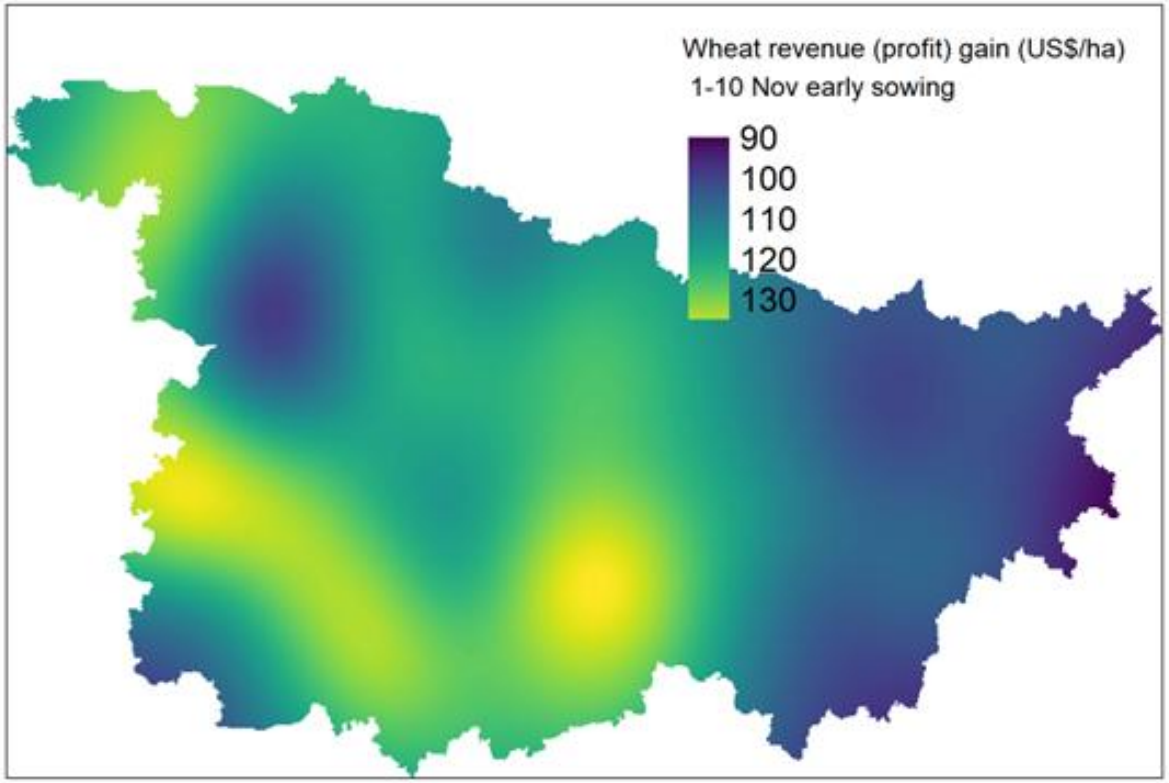
Predictors of best and worst farms (standardized)



Causal random forest spatial targeting using LCAS data for Bihar and Eastern UP

Ex ante farmer specific yield and profit gains to early sowing



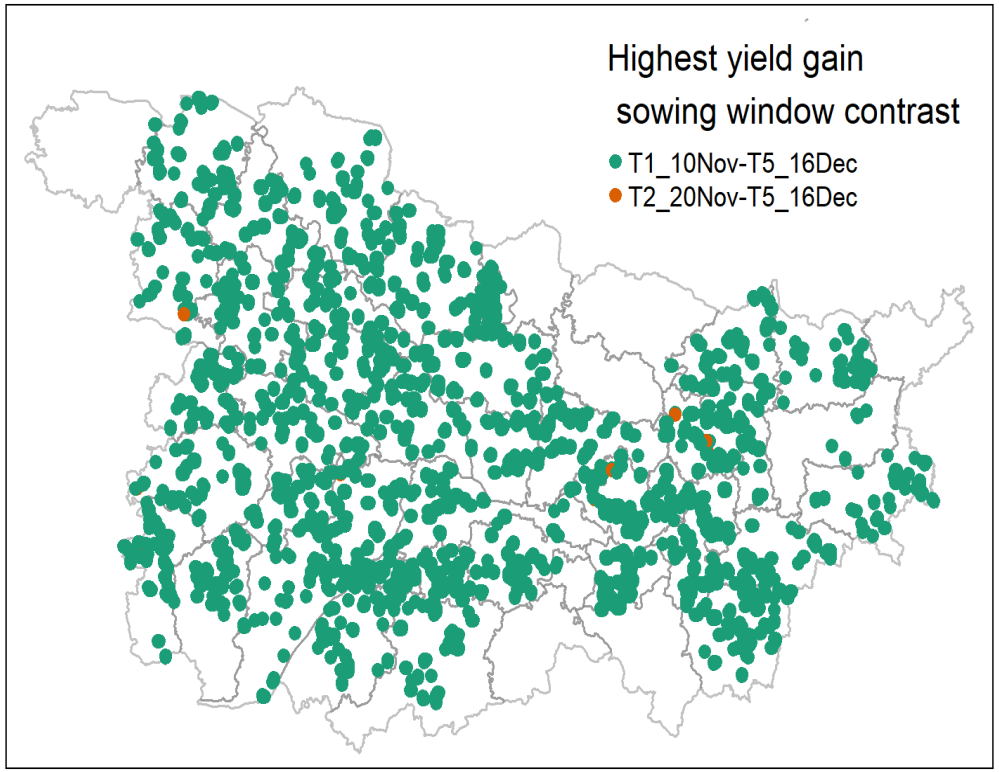


Mkondiwa et al 2025 forthcoming

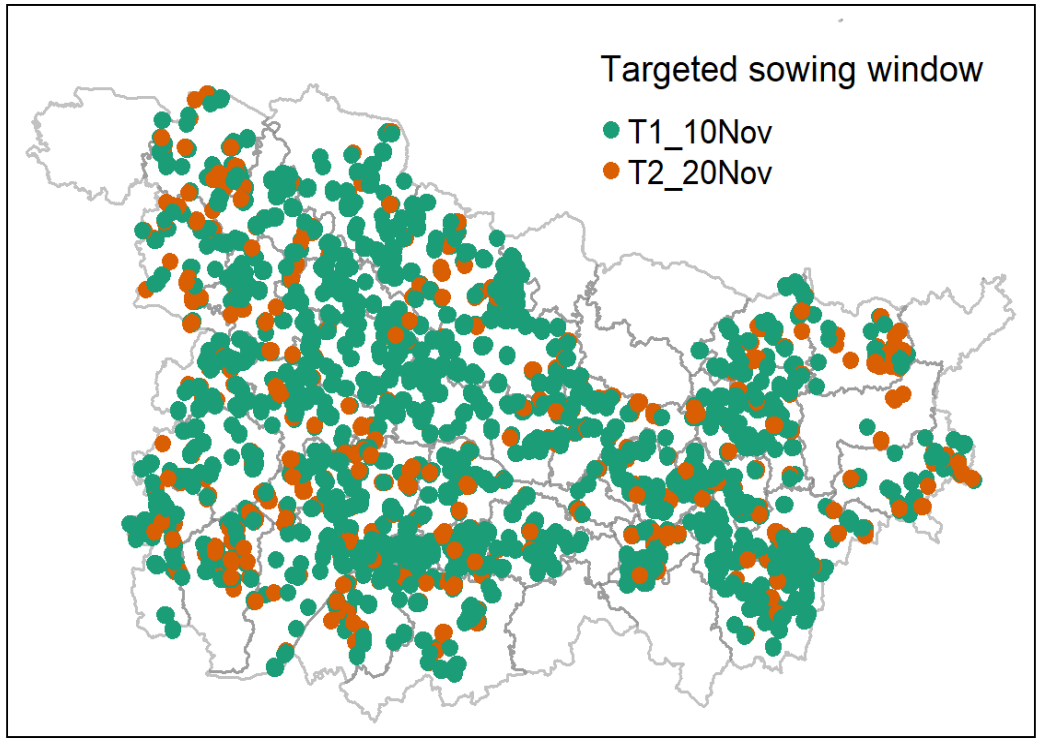
Full GRF vs policy tree targeting

Sowing dates

Full causal ML max yield option



Depth 2 [3 variables] max yield option



Session 3: Other Applications

Growing applications for agricultural policy analysis

nature food
ARTICLES
https://doi.org/10.1038/s43016-022-00493-z
Check for updates

Profitability of climate-smart soil fertility investment varies widely across sub-Saharan Africa

Ellen B. McCullough[1], Julianne D. Quinn[2] and Andrew M. Simons[3]

European Review of Agricultural Economics Vol 49 (4) (2022) pp. 723–759
doi:https://doi.org/10.1093/erae/jbab057
Advance Access Publication 9 March 2022

Using Machine Learning to Identify Heterogeneous Impacts of Agri-Environment Schemes in the EU: A Case Study

Christian Stetter*, Philipp Mennig and Johannes Sauer

Cameron-Harp *et al.*
Carbon Balance and Management (2024) 19:6
https://doi.org/10.1186/s13021-024-00252-6
Carbon Balance and Management
RESEARCH
Open Access
Check for updates

Predicting the spatial variation in cost-efficiency for agricultural greenhouse gas mitigation programs in the U.S.

Micah V. Cameron-Harp[1*], Nathan P. Hendricks[1] and Nicholas A. Potter[2]

ENVIRONMENTAL RESEARCH
LETTERS
LETTER • OPEN ACCESS

Combining randomized field experiments with observational satellite data to assess the benefits of crop rotations on yields

Dan M Kluger[3,1], Art B Owen[1] and David B Lobell[2]

Published 6 April 2022 • © 2022 The Author(s). Published by IOP Publishing Ltd
Environmental Research Letters, Volume 17, Number 4

Causal RF+ Policy learning

Targeting, Personalization, and Engagement in an Agricultural Advisory Service*

Susan Athey, Stanford GSB
Shawn Cole, Harvard University
Shanjukta Nath, University of Georgia
Jessica Zhu, PxD

August 9, 2023

DOI: 10.1111/1477-9552.12550
ORIGINAL ARTICLE
JAE Journal of Agricultural Economics

One size does not fit all: Heterogeneous economic impact of integrated pest management practices for mango fruit flies in Kenya—a machine learning approach

Kelvin Mulungu[1] | Zewdu Ayalew Abro[2] | Wambui Beatrice Muriithi[1] | Menale Kassie[1] | Miachael Kidoido[1] | Subramanian Sevgan[1] | Samira Mohamed[1] | Chrysantus Tanga[1] | Fathiya Khamis[1]

(1) Impact of 2022 drought on crop management decisions and rice yields in Bihar

- What is the impact of drought on crop management decisions and rice yields?
- For which type of farmers are these effects higher? By irrigation type, transplanting date, elevation,e.t.c.
- We used **causal random forest algorithm** to generate these heterogenous effects.

Agricultural Water Management 320 (2025) 109830

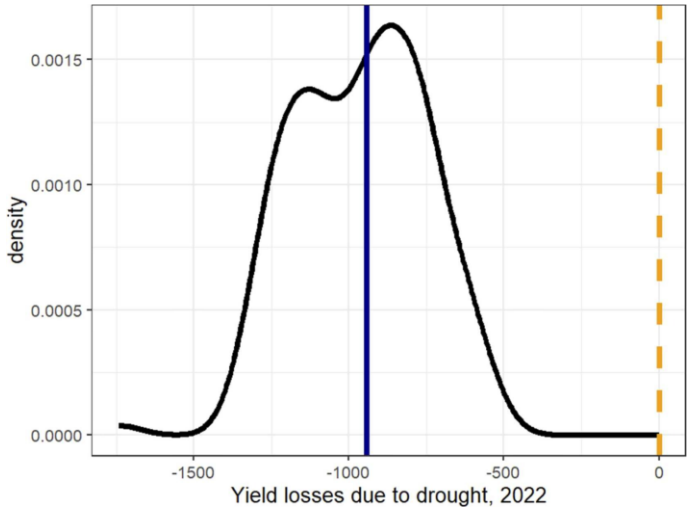
Contents lists available at ScienceDirect

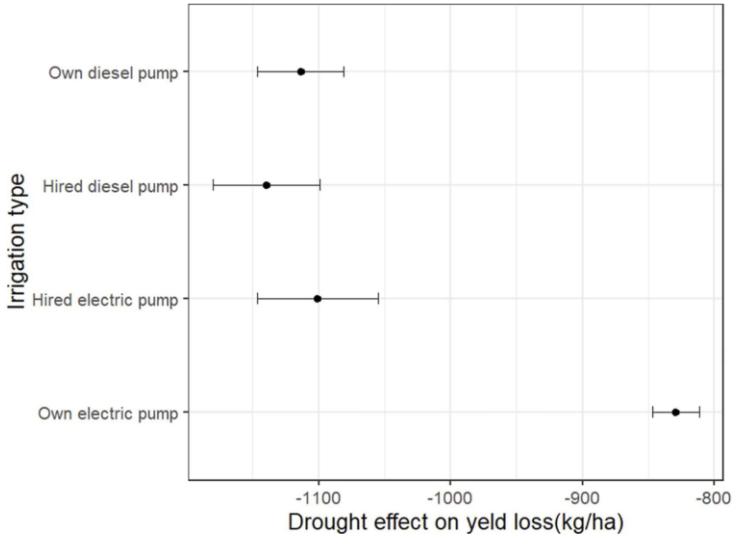
Agricultural Water Management

journal homepage: www.elsevier.com/locate/agwat

Farmers agronomic management responses to extreme drought and rice yields in Bihar, India

Maxwell Mkondiwa [a,*], Avinash Kishore [b], Prakashan Chellattan Veetil [c], Sonam Sherpa [d], Satyam Saxena [a], Bhavani Pinjarla [e], Anton Urfels [f,g,h], Shishpal Poonia [a], Anurag Ajay [a], Peter Craufurd [a], Ram Malik [a], Andrew McDonald [g]

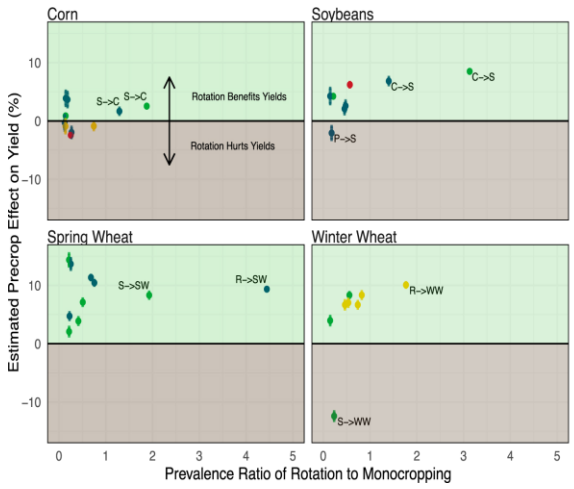


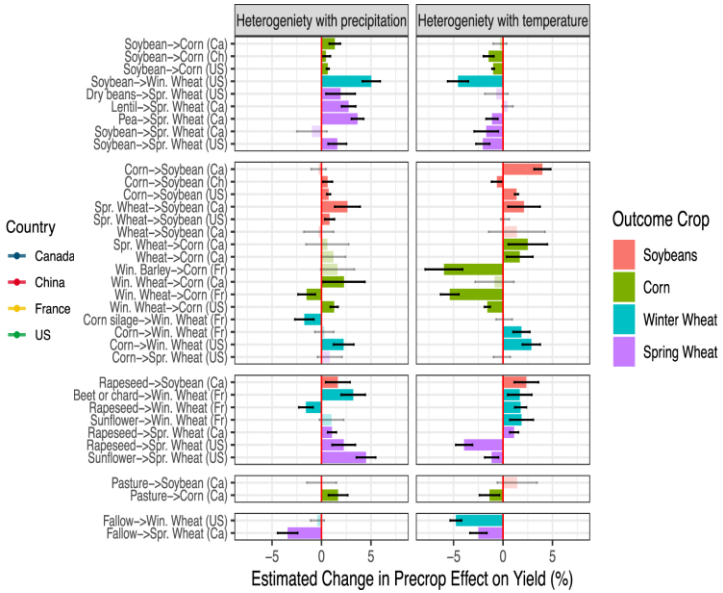


(2) Heterogeneous impacts of crop rotations on yields (or profits)

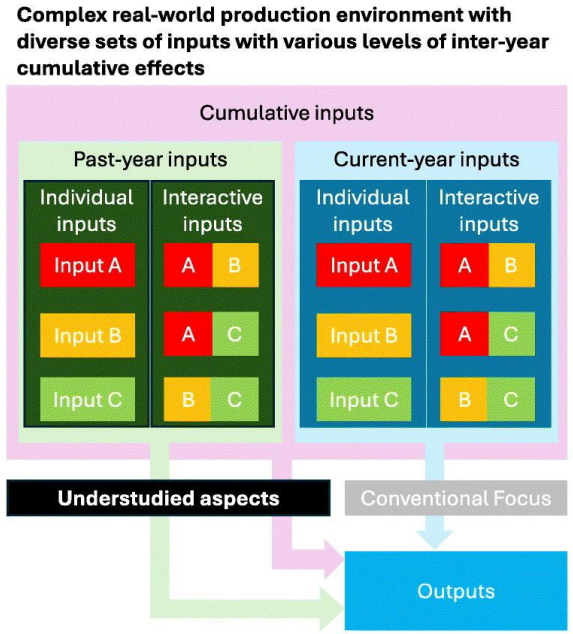
Causal random forest (Kluger et al 2025)

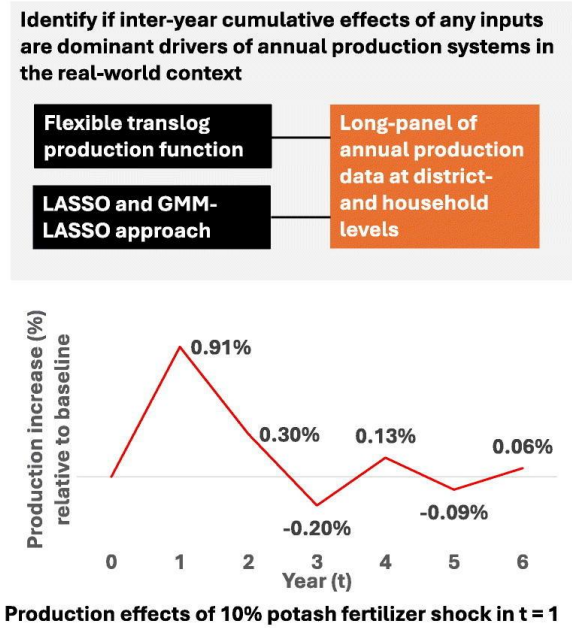
- How benefits of crop rotation vary for different crop sequences and weather conditions?





Post-Double LASSO (Takeshima and Kishore 2026)





(2) Impact assessment of government interventions and schemes, other applications

- **Aim:** To identify the environmental effectiveness of agri-environment schemes in European Union.

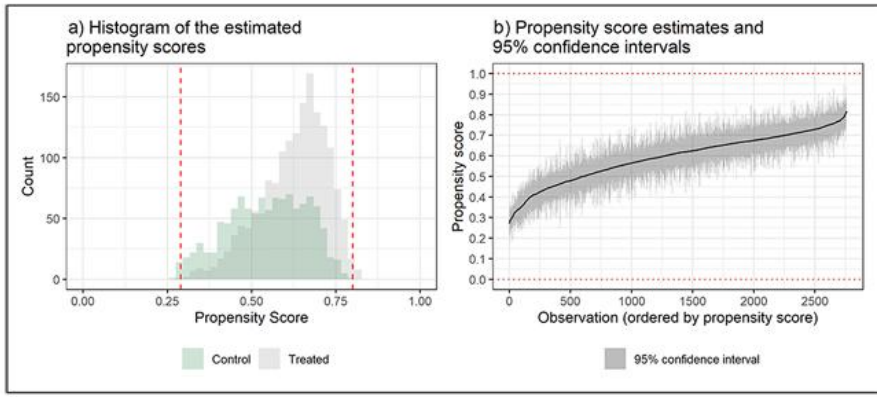


Fig. 2. Summary of the propensity scores obtained from the step-1 propensity forest

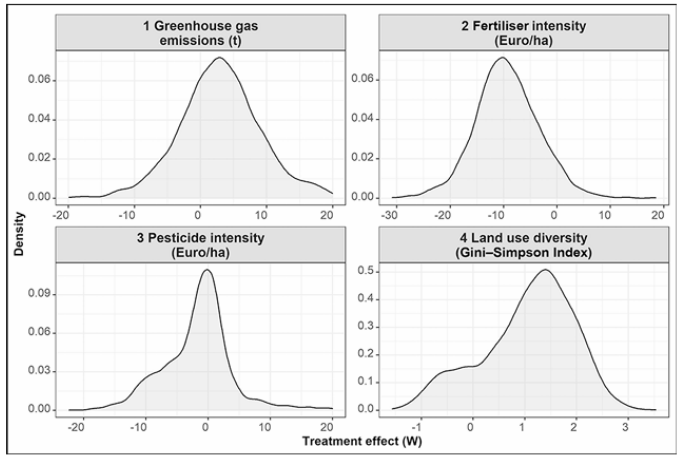


Fig. 3. Causal forest result: Distribution of the HTE estimates for the four environmental indicators.

Session 4: Policy learning and other advances

Culture 4: Decision and recommender systems

- Knowing personalized yield or profit gains does not translate to a policy decision
  - Should you target the highest yielding farmer? Lowest adoption farmers?
- Farmer specific, personalized, or context specific recommendations are increasingly possible.
  - Lessons from precision medicine/personalized medicine.
  - Lessons from tech marketing, e.g., Facebook, Amazon, Youtube.
- Personalized decision heuristics beyond **profit maximization to issues of risk aversion, fairness and other considerations.**

Minimal variables targeting: Optimal policy assignment

- Conventional approach to recommend optimal decisions is optimization
- That's assume a **functional form, estimate the parameters then choose optimal values at some ceteris paribus assumption**
- Causal ML allows
  - **Customization to subgroup or individuals: Treat if $\mathbf{E[\tau_i | X_i \in S] > 0}$**
  - Use policy learning by assuming a policy maker has a utilitarian regret objective function (Athey and Wager 2021)
  - **Policy:** Maps from observed features/characteristics to recommended actions

$$\hat{\pi} = argmax\left\{\hat{V}(\pi) : \pi \in \Pi\right\}$$

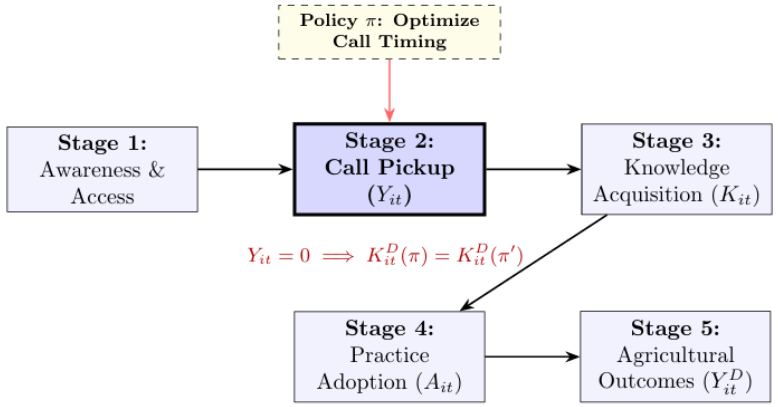
$$\hat{V}(\pi) = \frac{1}{n}\sum_{i=1}^{n}(2\pi(x_i) - 1)\,\hat{\Gamma}_i$$

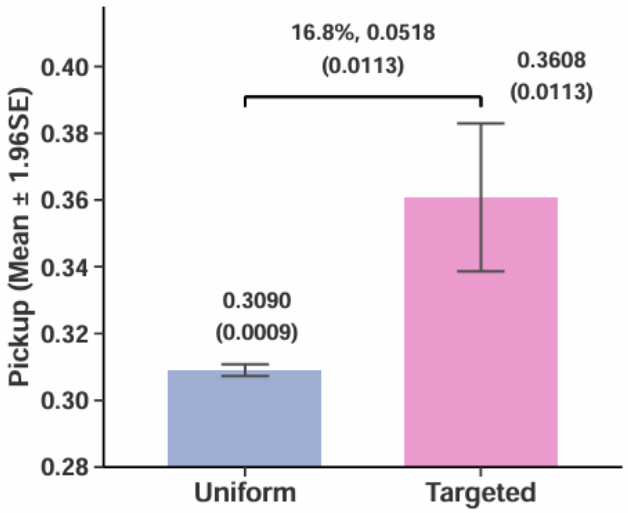
$$\hat{\Gamma}_i = \hat{\mu}_1(X_i) - \hat{\mu}_0(X_i) + \frac{W_i}{\hat{e}(X_i)}\left(Y_i - \mu_1(X_i)\right) - \frac{1 - W_i}{1 - \hat{e}(X_i)}\left(Y_i - \hat{\mu}_o(X_i)\right)$$

Example: Optimally targeting calls for agricultural advisory service in Odisha (Athey et al 2026)

- When should you call farmers to give agricultural advisory?
- Design an experiment
- Use LASSO and causal forests
- Then use policy learning to test if targeting the call reads higher pickup and yields

Figure 1: Marketing Funnel for Agricultural Advisory Service.





Culture 5: Equilibrium, system optimization and process understanding

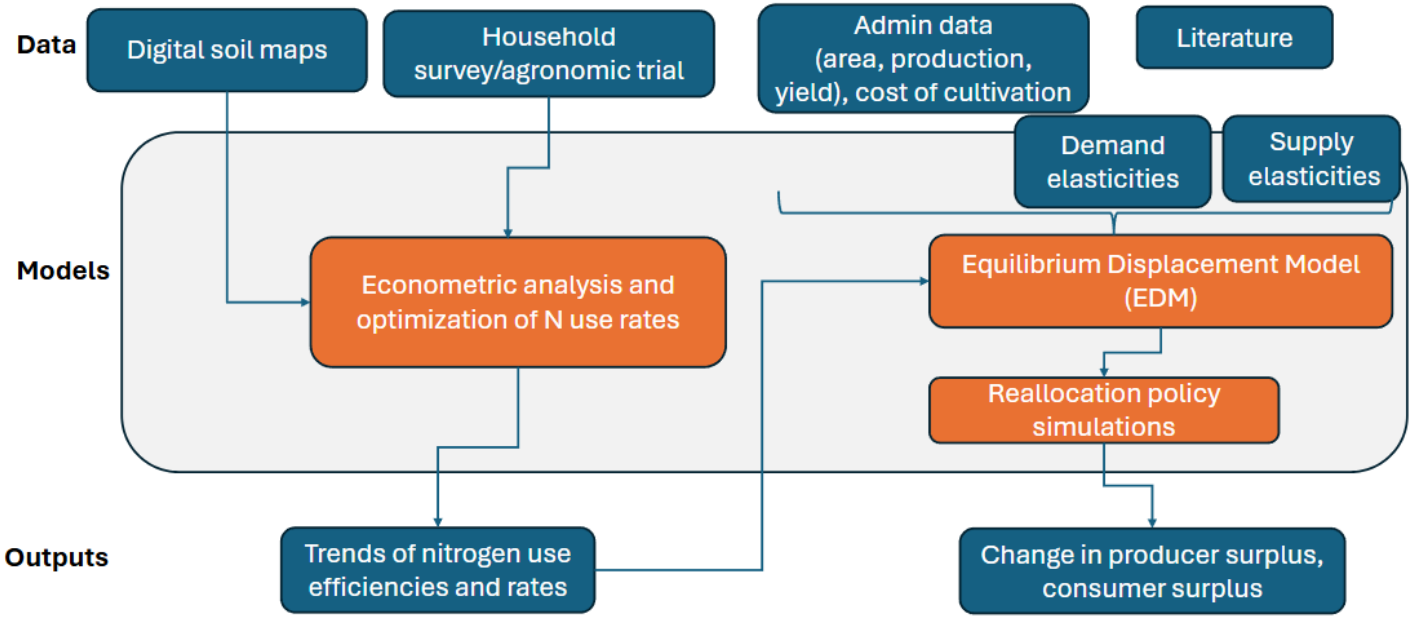
- Demand and supply thinking distinguishes us as economists.

- High production $\neq higher\ welfare$.

- Feedback mechanisms

Hybrid Causal AI + Economic Surplus Analysis

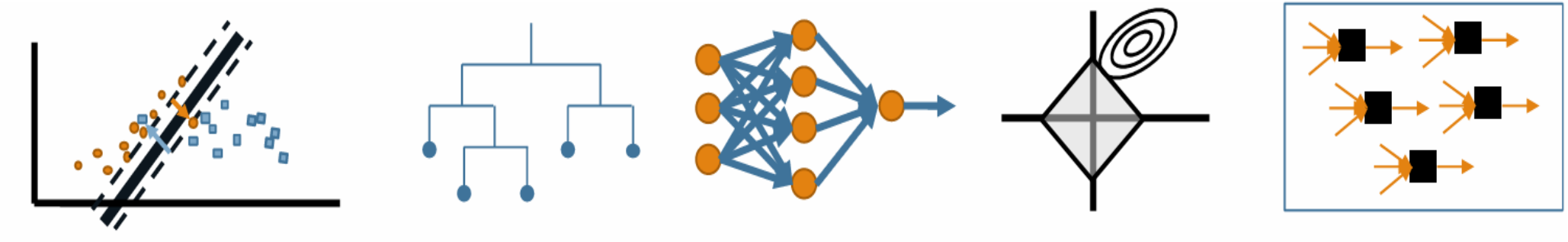
- **Spatially explicit economic surplus of nitrogen management in Tanzania.** **Combining causal AI and economic surplus approach** (Mkondiwa and Chamberlin 2024: Url: https://hdl.handle.net/10883/35304).



Causal ML

Economic Surplus Analysis

Summary: ML tools with a cause



- ML: Tools for learning to predict $y = f(x)$ when we don't know f
- Simple workflow across most causal ML tools
  - Predict propensity scores
  - Predict counterfactual outcomes by combining many controls
  - If you have too many variables, interactions, use LASSO
    - Get guidance on which (a) controls, or (b) instruments to include
    - Learn which instruments to use in IV
  - Learn form of heterogeneity in (a) error variance, and (b) treatment effects

If time permits:
Learning and reproducing repositories in the notes

- Use GitHub desktop

- Clone repositories

- Use Visual Studio Code
  - To easily use R, python and AI coding agents easily if needed

Thank you for your interest!

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